Cincinnati Children's Hospital Medical Center

Department of Pathology · Consultation Service

PATHOLOGY CONSULTATION REPORT

FINAL

Patient	SMITH, JANE	**Consult accession**	PV26-001-C
DOB / Age / Sex	03/14/2012 · 14 y · F	**Referring accession**	PV26-001 (Ohio State)
Received	10/09/2026 (digital)	**Final report**	10/14/2026
Specimen	Left forearm, soft tissue mass, excision	**Referring pathologist**	Swati Satturwar, MD

FINAL DIAGNOSIS

LEFT FOREARM, SOFT TISSUE MASS, EXCISION (REFERRING ACCESSION PV26-001):
– CIC-REARRANGED SARCOMA, WITH CIC::DUX4 FUSION DETECTED (SEE COMMENT).
– TUMOR SIZE: 3.1 CM. TUMOR IS 0.3 CM FROM THE DEEP INKED MARGIN (PER REFERRING REPORT).

COMMENT

Digital review of the referring slides shows a lobulated, high-grade round cell sarcoma with vesicular chromatin, conspicuous nucleoli, mild to moderate nuclear pleomorphism, myxoid stroma and geographic necrosis. The patchy CD99 and nuclear ETV4 and WT1 expression found at the referring institution, together with diffuse nuclear DUX4 staining performed here, are characteristic of CIC-rearranged sarcoma. EWSR1 break-apart FISH is negative, and RNA-based fusion sequencing identifies a CIC::DUX4 fusion, confirming the diagnosis.

CIC-rearranged sarcoma is classified separately from Ewing sarcoma (WHO Classification of Tumours, Soft Tissue and Bone Tumours, 5th edition). It typically follows a more aggressive clinical course and has been reported to respond less well to Ewing-type chemotherapy. Review at a pediatric sarcoma multidisciplinary tumor board is recommended.

The case was reviewed at our intradepartmental consensus conference, and the participating pathologists concur. The diagnosis was discussed with Dr. Satturwar on 10/12/2026, and the molecular result on 10/14/2026.

ANCILLARY STUDIES

Test	Performed at	Result
CD99 (IHC)	Ohio State	Positive; patchy, not diffuse membranous
ETV4 (IHC)	Ohio State	Positive; nuclear
WT1 (IHC)	Ohio State	Positive; nuclear
DUX4 (IHC)	Cincinnati Children's	Positive; diffuse, strong nuclear staining
EWSR1 break-apart FISH	Cincinnati Children's	Negative; no EWSR1 rearrangement detected
RNA fusion panel (NGS)	Cincinnati Children's	**CIC::DUX4 fusion detected**

DUX4 and molecular studies performed on block A3. Tissue for molecular testing was macrodissected from the tumor area annotated by the referring pathologist on the digital slide (estimated tumor content 80% after macrodissection). Tests were developed and their performance characteristics determined by the performing laboratory.

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CONSULTATION DETAILS

Referred by	Swati Satturwar, MD · Department of Pathology, The Ohio State University Wexner Medical Center
Question	Confirmation of suspected CIC-rearranged sarcoma; DUX4 immunohistochemistry and molecular testing.
Received	10/09/2026, digitally via PathPresenter (whole-slide images). Paraffin block A3 received 10/10/2026 for ancillary studies. No diagnostic glass slides were shipped.
Reviewed	H&E slides A1–A4; CD99, ETV4 and WT1 (block A3); referring preliminary report dated 10/09/2026; prior core biopsy PV26-000 (slides and report).

CLINICAL HISTORY (PER REFERRING INSTITUTION)

14-year-old girl with a slowly enlarging, painless nodule on the left forearm over six months. A prior core biopsy was inconclusive. Imaging was not available with the referral.

Archana Shenoy, MD
Pathologist, Cincinnati Children's Hospital Medical Center
Electronically signed: 10/14/2026 16:10 EDT

Final report returned electronically to the referring case (PV26-001) via PathPresenter; the referring pathologist was notified. Synoptic reporting and final margin assessment are deferred to the referring institution.